

Java Programming Assignment 3

Assignment Questions, Clarifications & Submission Guidelines

Question 1: User Input and File Handling

Write a Java program that demonstrates user input and file handling. The program should prompt the user to enter their complete name (the student should enter his/her first and last name) and a text file name, append the complete name to the specified file content, read the file's content, and display it on the console.

Clarifications

- Use the Scanner class to read user input for the file name.
- Append your complete name to the specified file, then read and display the file's content on the console.
- Handle any possible errors, such as file not found or input mismatch exceptions.

Question 2: Exception Handling

Write a Java program that demonstrates the use of exceptions for error handling. The program should handle a divide by zero exception and display an appropriate message.

Clarifications

- Implement try-catch blocks to handle exceptions.
- Simulate a division by zero operation and catch the ArithmeticException.
- Display 'Division by zero is not possible' if a division by zero occurs.

Expected Output

```
Enter numerator: 10
Enter denominator: 0
Division by zero is not possible.
```

Question 3: BufferedReader

Write a Java program that demonstrates BufferedReader for reading input from the user. The program should read a user's name and age, then display them in a message.

Clarifications

- Use BufferedReader to read user input for name and age.
- Use the readLine() method to read strings and the parseInt() method to read integers.
- Display the user's name and age in a formatted message.

Expected Output

```
Enter your name: John Doe
Enter your age: 30
Hello, John Doe! You are 30 years old.
```

Question 4: Collections Framework

Write a Java program that demonstrates the use of ArrayLists, LinkedLists, and HashSet from the Collections Framework. The program should add elements to each of these collections and display them.

Clarifications

- Create an ArrayList and add a few elements. Display the ArrayList.
- Create a LinkedList, add elements, and display the LinkedList.
- Create a HashSet, add elements, and display the HashSet.
- Explain the differences between ArrayList, LinkedList, and HashSet.

Collection	Main characteristic
ArrayList	Fast access using index; maintains insertion order.
LinkedList	Suitable for insertion/removal operations; maintains order.
HashSet	Does not allow duplicate elements; order is not guaranteed.

Simple Submission Guidelines

1. Create a folder on your computer and name it according to the assignment number, for example: Assignment 3.
2. Take full-screen screenshots of each task.
3. Each screenshot must clearly show the complete code, output, and IDE.
4. Do not use online compiler screenshots.
5. Do not crop or edit the screenshots.
6. Add all screenshots inside the assignment folder.
7. Convert the folder into a ZIP file.
8. Upload the ZIP file in the assignment submission section.
9. Submit the assignment after uploading the ZIP file successfully.

Strict Notice on Course Project Submissions

- All learners are mandatorily required to complete the course lectures and implement the projects exactly as demonstrated by the course instructor during the lectures.
- This includes the same coding approach and logic, structure, flow, methodology, concepts, syntax usage, implementation style, steps, and techniques taught by the instructor.
- The use of AI tools, code generators, or externally sourced code (including partial assistance) is strictly prohibited.
- Submissions must reflect only what has been taught and implemented by the instructor in the course lectures.
- Any submission found to be generated using AI, contain logic or structure not covered in the lectures, or differ from the instructor's demonstrated implementation may be considered incorrect and irrelevant.
- Such submissions may be rejected without evaluation, and continued non-compliance may affect course completion and certificate eligibility.

Important: Write the Java implementations yourself using the exact approach taught in your course lectures, and add your own full-screen code/output screenshots before submitting.